

PROBABILITY

Complete Case-by-Case Study Guide

Definitions, OR/AND rules, overlap, replacement, conditional probability, independence, and Bayes' theorem, each illustrated with coin, card, dice, and e-commerce examples.

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1. Probability Vocabulary and Notation

An outcome is one possible result of an experiment. The sample space, denoted S , is the set of all possible outcomes. An event is a collection of outcomes that satisfies a condition.

Notation	Meaning
A	Event A
A'	A does not occur; complement of A
A and B	Both A and B occur
A or B	The OR event: an outcome satisfying A or B
$P(A)$	Probability of A
$P(A \text{ and } B)$	Probability that both A and B occur
$P(A \text{ or } B)$	Probability of the OR event
$P(A \text{ given } B)$	Probability of A when B is known to have occurred

$P(A)$ = favorable outcomes / total outcomes

Example — Dice

Example: A spinner is divided into 4 equal sections labeled 1, 2, 3, 4. Let A = spinning an odd number. What is the sample space and $P(A)$?

1. The sample space is every possible section: $S = \{1, 2, 3, 4\}$.
2. $A = \{1, 3\}$, since those are the odd numbers.
3. $P(A) = \text{favorable} / \text{total} = 2/4$.

Answer: $P(A) = 2/4 = 1/2$.

Example — E-Commerce

Example: An online store logs each visit as either 'browsed only', 'added to cart', or 'purchased'. Let A = a visit that ends in a purchase. What is the sample space, and how does A relate to it?

1. The sample space is every possible outcome of a visit: $S = \{\text{browsed only, added to cart, purchased}\}$.
2. $A = \{\text{purchased}\}$ is the single outcome of interest within S .
3. This shows an event doesn't need to contain multiple outcomes — a single specific result is still a valid event.

Answer: $S = \{\text{browsed only, added to cart, purchased}\}$; $A = \{\text{purchased}\}$.

2. Basic Probability

For equally likely outcomes, probability is the number of favorable outcomes divided by the total number of possible outcomes.

$P(A)$ = number of outcomes in A / number of outcomes in S

Probabilities always satisfy $0 \leq P(A) \leq 1$. A probability of 0 means the event is impossible; a probability of 1 means the event is certain.

Example — Coins

Example: A fair coin is tossed once. What is the probability of getting tails?

1. There is 1 favorable outcome: tails.
2. There are 2 total outcomes: heads or tails.
3. Probability = $1/2$.

Answer: $P(\text{tails}) = 1/2$.

Example — E-Commerce

Example: An online store had 500 visitors yesterday, and 60 of them completed a purchase. If a visitor is chosen at random from yesterday's log, what is the probability that visitor made a purchase?

1. Total visitors = 500.
2. Visitors who purchased = 60.
3. Probability = $60/500 = 0.12$.

Answer: $P(\text{purchase}) = 0.12$, or 12%.

3. Complement

The complement of A is the event that A does not occur.

$$P(A') = 1 - P(A) \quad P(A) = 1 - P(A')$$

Example — Dice

Example: A fair die is rolled once. What is the probability of NOT rolling a 1 or a 2?

1. $P(\text{rolling 1 or 2}) = 2/6$, since two of the six faces satisfy the condition.
2. Use the complement: $P(\text{not 1 or 2}) = 1 - 2/6$.
3. $1 - 2/6 = 4/6 = 2/3$.

Answer: $P(\text{not 1 or 2}) = 2/3$.

Example — E-Commerce

Example: Historical data shows that 70% of online shopping carts are abandoned before checkout. What is the probability that a given cart is completed (checked out)?

1. 'Completed' is the complement of 'abandoned'.
2. $P(\text{abandoned}) = 0.70$.
3. $P(\text{completed}) = 1 - 0.70 = 0.30$.

Answer: $P(\text{cart completed}) = 0.30$, or 30%.

Useful Shortcut: At Least One

If a question asks for 'at least one' and it is easier to calculate 'none', use the complement: $P(\text{at least one}) = 1 - P(\text{none})$.

Example — Dice: A fair die is rolled 4 times. What is the probability of rolling at least one 6?

1. 'At least one' is easiest through the complement: at least one 6 = $1 - P(\text{no 6 in 4 rolls})$.
2. $P(\text{no 6 on a single roll}) = 5/6$.
3. Since the rolls are independent, $P(\text{no 6 in 4 rolls}) = (5/6)^4 \approx 0.4823$.
4. Subtract from 1: $1 - 0.4823 = 0.5177$.

Answer: $P(\text{at least one 6}) \approx 0.518$, or about 51.8%.

4. OR and the Addition Rule

A OR B means either A or B happens. The OR event collects the outcomes that satisfy A or B. If A and B can occur together, the shared outcomes are included once in the final OR event.

A AND B means both A and B happen. When A and B can happen together, A AND B is the overlap between the two events.

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Subtracting $P(A \text{ and } B)$ avoids double-counting: if an outcome belongs to both A and B, adding $P(A)$ and $P(B)$ counts it twice, so the overlap is subtracted once to leave it counted exactly once.

Example — Cards, With Overlap

Example: One card is drawn from a standard 52-card deck. What is $P(\text{black card or face card})$?

1. Let A = black card. There are 26 black cards, so $P(A) = 26/52$.
2. Let B = face card (jack, queen, king). There are 12 face cards, so $P(B) = 12/52$.
3. A and B overlap: black face cards are the 6 black jacks, queens, and kings, so $P(A \text{ and } B) = 6/52$.
4. $P(A \text{ or } B) = 26/52 + 12/52 - 6/52 = 32/52 = 8/13$.

Answer: $P(\text{black or face card}) = 8/13$.

Example — Dice, Mutually Exclusive

Example: A fair die is rolled. What is $P(\text{rolling a 1 or a 6})$?

1. A = rolling 1 and B = rolling 6.
2. These outcomes cannot happen on the same roll, so $P(A \text{ and } B) = 0$.
3. $P(A \text{ or } B) = 1/6 + 1/6 = 2/6 = 1/3$.

Answer: $P(1 \text{ or } 6) = 1/3$.

Example — E-Commerce

Example: Out of 1,000 subscribers, 300 opened a marketing email, 150 clicked on an ad, and 50 did both. What is the probability that a randomly chosen subscriber opened the email OR clicked the ad?

1. Let A = opened email, so $P(A) = 300/1000$.
2. Let B = clicked ad, so $P(B) = 150/1000$.
3. The overlap (did both) is $P(A \text{ and } B) = 50/1000$.
4. $P(A \text{ or } B) = 300/1000 + 150/1000 - 50/1000 = 400/1000 = 2/5$.

Answer: $P(\text{opened email or clicked ad}) = 2/5$, or 40%.

Three OR Cases to Recognize

Case	Can both happen?	Rule
Overlapping A and B	Yes	$P(A)+P(B)-P(A \text{ and } B)$
Mutually exclusive A and B	No	$P(A)+P(B)$
A and B are the same event	Yes, completely	$P(A \text{ or } B)=P(A)$

5. AND and the Multiplication Rule

A AND B means both A and B happen. The multiplication rule connects an AND probability to a conditional probability.

$$P(A \text{ and } B) = P(A) \times P(B \text{ given } A) \quad \text{equivalently} \quad P(A \text{ and } B) = P(B) \times P(A \text{ given } B)$$

Important: the word AND only tells you that both conditions must occur. It does not tell you whether the events are independent. For an AND question, first ask whether the probability of the second event changes after the first event occurs.

Example — Coin and Die

Example: A fair coin is tossed and a fair die is rolled at the same time. What is the probability of getting heads AND rolling a 4?

1. $P(\text{heads}) = 1/2$.
2. The coin and die don't influence each other, so $P(\text{rolling } 4 \text{ given heads}) = 1/6$.
3. Multiply: $1/2 \times 1/6 = 1/12 \approx 0.083$.

Answer: $P(\text{heads and } 4) = 1/12$.

Example — E-Commerce

Example: 60% of visitors to a shopping site browse on mobile devices. Among mobile browsers, 5% go on to make a purchase. What is the probability that a random visitor browses on mobile AND makes a purchase?

1. $P(\text{mobile}) = 0.60$.
2. $P(\text{purchase given mobile}) = 0.05$.
3. Multiply: $0.60 \times 0.05 = 0.03$.

Answer: $P(\text{mobile and purchase}) = 0.03$, or 3%.

6. With Replacement

With replacement means an item is returned to the population after each selection. The total number and composition return to the original state, so each draw has the same probabilities.

Example — Dice

Example: A fair die is rolled twice. What is the probability of rolling a 5 both times?

1. Each roll is independent since the die 'resets' automatically — rolling doesn't remove faces.
2. $P(\text{first roll is 5}) = 1/6$.
3. $P(\text{second roll is 5}) = 1/6$, unaffected by the first roll.
4. Multiply: $1/6 \times 1/6 = 1/36$.

Answer: $P(5 \text{ then } 5) = 1/36$.

Example — E-Commerce

Example: A loyalty database has 3 gold-tier and 7 regular-tier customers. For a promotional drawing, a name is drawn, recorded, and put back before the next drawing (with replacement). What is the probability that two draws in a row both select a gold-tier customer?

1. $P(\text{first draw gold}) = 3/10$.
2. Since the name is replaced, the pool resets to 3 gold and 7 regular.
3. $P(\text{second draw gold}) = 3/10$.
4. Multiply: $3/10 \times 3/10 = 9/100 = 0.09$.

Answer: $P(\text{gold, then gold}) = 0.09$, or 9%.

Replacement Comparison

Question	With replacement	Without replacement
Red then red (3 red, 2 blue)	$3/5 \times 3/5 = 9/25$	$3/5 \times 2/4 = 3/10$
Red then blue	$3/5 \times 2/5 = 6/25$	$3/5 \times 2/4 = 3/10$
Blue then red	$2/5 \times 3/5 = 6/25$	$2/5 \times 3/4 = 3/10$

7. Without Replacement

Without replacement means the selected item is not returned. The total number and/or composition changes after the first selection, so the second probability is usually dependent on the first.

Example — Cards

Example: Two cards are drawn from a standard 52-card deck without replacement. What is the probability that both cards are red?

1. $P(\text{first red}) = 26/52$.
2. After removing one red card, 25 red remain among 51 total cards.
3. $P(\text{second red given first red}) = 25/51$.
4. Multiply: $26/52 \times 25/51 = 650/2652 = 25/102 \approx 0.245$.

Answer: $P(\text{both red}) = 25/102 \approx 0.245$.

Example — E-Commerce

Example: A shipment of 20 units contains 3 defective units. Quality control inspects 2 units at random without replacement. What is the probability that both inspected units are defective?

1. $P(\text{first defective}) = 3/20$.
2. After removing one defective unit, 2 defective units remain among 19 total.
3. $P(\text{second defective given first defective}) = 2/19$.
4. Multiply: $3/20 \times 2/19 = 6/380 = 3/190 \approx 0.0158$.

Answer: $P(\text{both defective}) = 3/190 \approx 1.58\%$.

8. Conditional Probability

Conditional probability answers: 'What is the probability of A when we already know that B has happened?' The condition after 'given' becomes the restricted group to work inside.

$$P(\text{A given B}) = P(\text{A and B}) / P(\text{B})$$

Example — Dice

Example: A fair die is rolled. Given that the result is greater than 2, what is the probability that it is also even?

1. The condition restricts the sample space to {3, 4, 5, 6}, which has 4 outcomes.
2. Within that restricted group, the even outcomes are {4, 6}, which is 2 outcomes.
3. $P(\text{even given greater than 2}) = 2/4 = 1/2$.

Answer: $P(\text{even given greater than 2}) = 1/2$.

Example — E-Commerce

Example: Of 200 customers, 120 are on a premium subscription plan. Of those 120 premium customers, 84 renewed their subscription this year. What is $P(\text{renewed given premium})$?

1. The condition is 'premium', so restrict attention to the 120 premium customers.
2. Of those 120, 84 renewed.
3. $P(\text{renewed given premium}) = 84/120 = 0.7$.

Answer: $P(\text{renewed given premium}) = 0.7$, or 70%.

9. AND + Conditional Probability

For a dependent AND problem, calculate the first probability and then the conditional probability of the second event. The second probability is not necessarily the original probability, because the first event may change what remains.

$$P(\text{A and B}) = P(\text{A}) \times P(\text{B given A})$$

Example — Cards

Example: Two cards are drawn without replacement from a standard deck. What is the probability that the first is an ace AND the second is a king?

1. $P(\text{first ace}) = 4/52$.
2. Removing an ace doesn't affect the number of kings, so 4 kings remain among 51 cards.
3. $P(\text{second king given first ace}) = 4/51$.
4. Multiply: $4/52 \times 4/51 = 16/2652 = 4/663 \approx 0.006$.

Answer: $P(\text{ace then king}) = 4/663$.

Example — E-Commerce

Example: A cart contains 10 items: 8 are in stock and 2 are flagged as out-of-stock. Two items are randomly selected without replacement for a shipping check. What is the probability that both selected items are out-of-stock?

1. $P(\text{first out-of-stock}) = 2/10$.
2. After removing one out-of-stock item, 1 out-of-stock item remains among 9 total.
3. $P(\text{second out-of-stock given first out-of-stock}) = 1/9$.
4. Multiply: $2/10 \times 1/9 = 2/90 = 1/45 \approx 0.022$.

Answer: $P(\text{both out-of-stock}) = 1/45 \approx 2.2\%$.

10. Independence

A and B are independent when knowing that B occurred does not change the probability of A, and knowing that A occurred does not change the probability of B.

If A and B are independent: $P(A \text{ given } B) = P(A)$; $P(B \text{ given } A) = P(B)$; $P(A \text{ and } B) = P(A) \times P(B)$

A common test for independence: if $P(A \text{ and } B) = P(A) \times P(B)$, then A and B are independent.

Example — Coin and Die

Example: A fair coin is tossed and a fair die is rolled. Verify that getting heads and rolling a 3 are independent events.

1. $P(\text{heads}) = 1/2$ and $P(\text{rolling } 3) = 1/6$.
2. If independent, $P(\text{heads and } 3)$ should equal $P(\text{heads}) \times P(3) = 1/2 \times 1/6 = 1/12$.
3. Since the coin and die do not physically interact, $P(\text{heads and } 3)$ is indeed $1/12$, matching the product.

Answer: Yes — heads and rolling a 3 are independent, since $P(A \text{ and } B) = P(A) \times P(B)$.

Example — E-Commerce

Example: A store checks whether device type and purchase behavior are independent. $P(\text{mobile visitor}) = 0.5$ and $P(\text{purchase}) = 0.1$. Actual data shows $P(\text{mobile and purchase}) = 0.03$. Are the two events independent?

1. If independent, $P(\text{mobile and purchase})$ should equal $P(\text{mobile}) \times P(\text{purchase}) = 0.5 \times 0.1 = 0.05$.
2. The actual observed value is 0.03, not 0.05.
3. Since $0.03 \neq 0.05$, the events are not independent.

Answer: No — mobile browsing and purchasing are not independent; mobile visitors purchase less often than independence would predict.

11. Mutually Exclusive Versus Independent

These concepts answer different questions. Mutually exclusive events cannot occur together ($P(A \text{ and } B) = 0$). Independent events can both occur, but neither changes the probability of the other.

Concept	Meaning	AND probability	Can both occur?
Mutually exclusive	They cannot occur together	$P(A \text{ and } B) = 0$	No
Independent	One does not change the probability of the other	$P(A \text{ and } B) = P(A) \times P(B)$	Yes, if both have positive probability

Two events with positive probabilities cannot be both mutually exclusive and independent. Mutually exclusive means the overlap is zero; independence means the occurrence of one does not change the probability of the other.

Example — Dice

Example: A fair die is rolled. Let A = rolling less than 3 ($\{1, 2\}$) and B = rolling greater than 4 ($\{5, 6\}$). Are A and B mutually exclusive? Are they independent?

1. A and B share no outcomes, so $P(A \text{ and } B) = 0$ — they are mutually exclusive.
2. Check independence: $P(A) \times P(B) = (2/6) \times (2/6) = 4/36 = 1/9$, which is not 0.
3. Since $P(A \text{ and } B) \neq P(A) \times P(B)$, the events are not independent.

Answer: Mutually exclusive: yes. Independent: no.

Example — E-Commerce

Example: For a single order, let A = the order ships to the US, and B = the order ships to the UK. Are A and B mutually exclusive? Could they be independent?

1. A single order has exactly one shipping destination, so A and B cannot both occur: $P(A \text{ and } B) = 0$. They are mutually exclusive.
2. For independence we'd need $P(A \text{ and } B) = P(A) \times P(B)$. Since $P(A)$ and $P(B)$ are both positive but their product can't be 0, the equality fails.
3. This illustrates the general rule: two events with positive probabilities can never be both mutually exclusive and independent.

Answer: Mutually exclusive: yes. Independent: no.

12. Bayes' Theorem

Bayes' theorem reverses a conditional probability. It is especially useful when the known information is about a result and the question asks about its possible cause.

$$P(A \text{ given } B) = P(B \text{ given } A) \times P(A) / P(B)$$

Example — Medical Test (Full Numerical Walkthrough)

Example: A disease affects 1% of a population. A test is positive for 95% of people with the disease. Among people without the disease, 10% still test positive. If a randomly selected person tests positive, what is the probability that the person actually has the disease?

1. $P(\text{disease}) = 0.01$, so $P(\text{not disease}) = 0.99$.
2. $P(\text{positive given disease}) = 0.95$ and $P(\text{positive given not disease}) = 0.10$.
3. Find $P(\text{positive}) = (0.95 \times 0.01) + (0.10 \times 0.99) = 0.0095 + 0.099 = 0.1085$.
4. Apply Bayes: $P(\text{disease given positive}) = (0.95 \times 0.01) / 0.1085 \approx 0.0876$.

Answer: About 8.76%.

Example — Coins

Example: A box has two coins: a fair coin ($P(\text{heads}) = 0.5$) and a biased coin ($P(\text{heads}) = 0.9$). A coin is picked at random and flipped, landing heads. What is the probability the fair coin was picked?

1. $P(\text{fair}) = 0.5$ and $P(\text{biased}) = 0.5$, since the coin is picked at random.
2. $P(\text{heads given fair}) = 0.5$ and $P(\text{heads given biased}) = 0.9$.
3. Find $P(\text{heads}) = (0.5 \times 0.5) + (0.9 \times 0.5) = 0.25 + 0.45 = 0.70$.
4. Apply Bayes: $P(\text{fair given heads}) = (0.5 \times 0.5) / 0.70 = 0.25 / 0.70 \approx 0.357$.

Answer: $P(\text{fair given heads}) \approx 0.357$, or about 35.7%.

Example — E-Commerce

Example: 5% of an online store's orders are fraudulent. A fraud detector correctly flags 98% of actual fraudulent orders, but also incorrectly flags 2% of legitimate orders. If an order is flagged, what is the probability it is actually fraudulent?

1. $P(\text{fraud}) = 0.05$, so $P(\text{legitimate}) = 0.95$.
2. $P(\text{flagged given fraud}) = 0.98$ and $P(\text{flagged given legitimate}) = 0.02$.
3. Find $P(\text{flagged}) = (0.98 \times 0.05) + (0.02 \times 0.95) = 0.049 + 0.019 = 0.068$.
4. Apply Bayes: $P(\text{fraud given flagged}) = (0.98 \times 0.05) / 0.068 = 0.049 / 0.068 \approx 0.721$.

Answer: $P(\text{fraud given flagged}) \approx 0.721$, or about 72.1%.

13. Mixed Case Selection Guide

Before calculating, identify the structure of the question using the key wording.

Wording / clue	Likely structure	First formula to consider
'AND', 'both', 'then'	Multiplication	$P(A \text{ and } B) = P(A) \times P(B \text{ given } A)$
'OR', 'either'	Addition	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
'given', 'among those who'	Conditional	$P(A \text{ given } B) = P(A \text{ and } B) / P(B)$
'not', 'does not'	Complement	$P(A') = 1 - P(A)$
'with replacement'	Usually same probabilities	Check whether probabilities reset
'without replacement'	Usually dependent	Update the second probability
'independent'	No probability change	$P(A \text{ and } B) = P(A) \times P(B)$
'mutually exclusive'	No overlap	$P(A \text{ and } B) = 0$

Decision Sequence

- Identify the events A and B.
- Circle the key wording: OR, AND, given, not, replacement, independent, mutually exclusive.
- If OR: ask whether A and B overlap.
- If AND: ask whether the second probability changes after the first.
- If 'given' appears: restrict the sample space to the condition.
- If 'without replacement': update the counts after the first selection.
- If 'with replacement': return the item and restore the original counts.
- Check that the final probability is between 0 and 1.

14. Common Traps and Wording

Trap 1: Treating every OR question as mutually exclusive

Wrong approach: $P(A \text{ or } B) = P(A) + P(B)$ every time. Correct approach: use the general rule first; the formula simplifies only when the overlap is actually zero.

Trap 2: Calling the overlap itself an OR outcome

The overlap is A AND B. It is included inside the OR event when the OR event is formed — it is not itself a separate 'OR' outcome.

Trap 3: Assuming AND means independent

AND only means both occur. Independence must be established separately by checking whether $P(A \text{ and } B) = P(A) \times P(B)$.

Trap 4: Forgetting the denominator after 'given'

$P(A \text{ given } B)$ is not $P(A \text{ and } B)$ divided by the total population. It is $P(A \text{ and } B)$ divided by $P(B)$.

Trap 5: Forgetting the population changes

Without replacement, the second probability may change. Always update both the numerator and denominator after each selection.

Trap 6: Using Bayes when ordinary conditional probability is enough

Bayes is useful specifically when you know $P(B \text{ given } A)$ but need $P(A \text{ given } B)$. Don't reach for it just because the word 'probability' appears.

15. Worked Mixed Practice

#	Problem	Answer
1	P(even or greater than 4) on one die	2/3
2	P(2 or 5) on one die	1/3
3	Both cards aces, without replacement	1/221
4	Both cards aces, with replacement	1/169
5	P(female given passed), class of 25	3/5
6	P(A)=0.3, P(B)=0.4 independent, find P(A and B)	0.12
7	P(A)=0.6, P(B)=0.5, P(A and B)=0.2, find P(A or B)	0.9
8	P(A)=0.6, P(B)=0.5 mutually exclusive, find P(A or B)	Inconsistent (sum exceeds 1)

Additional Practice — Cards/Dice

Practice A: A committee of 2 people is chosen at random, without replacement, from a group of 5 men and 3 women. What is the probability both chosen are women?

1. P(first woman) = $3/8$.
2. After removing one woman, 2 women remain among 7 people.
3. P(second woman given first woman) = $2/7$.
4. Multiply: $3/8 \times 2/7 = 6/56 = 3/28$.

Answer: P(both women) = $3/28$.

Additional Practice — E-Commerce

Practice B: A factory's machine produces defective items 2% of the time. If two items are inspected independently, what is the probability that at least one is defective?

1. Use the complement: at least one defective = $1 - P(\text{neither defective})$.
2. P(one item is not defective) = 0.98.
3. Since the items are independent, $P(\text{neither defective}) = 0.98 \times 0.98 = 0.9604$.
4. Subtract from 1: $1 - 0.9604 = 0.0396$.

Answer: P(at least one defective) = 0.0396, or about 3.96%.

Practice C: An e-commerce site finds that 25% of customers who abandon a cart return within a week to complete the purchase. Of 400 abandoned carts last month, how many would you expect to be completed within a week, and what's the probability a random abandoned cart is NOT completed?

1. Expected completions = $400 \times 0.25 = 100$ carts.
2. P(not completed) is the complement of P(completed).
3. P(not completed) = $1 - 0.25 = 0.75$.

Answer: Expected completions = 100 carts; P(not completed within a week) = 0.75.

16. Final Formula Sheet and Exam Checklist

Basic probability:

$P(A)$ = favorable outcomes / total outcomes

Complement:

$$P(A') = 1 - P(A)$$

OR — general addition rule:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

OR — mutually exclusive special case:

$$P(A \text{ and } B) = 0, \text{ so } P(A \text{ or } B) = P(A) + P(B)$$

AND — multiplication rule:

$$P(A \text{ and } B) = P(A) \times P(B \text{ given } A) = P(B) \times P(A \text{ given } B)$$

Conditional probability:

$$P(A \text{ given } B) = P(A \text{ and } B) / P(B)$$

Independence:

$$P(A \text{ given } B) = P(A); \quad P(A \text{ and } B) = P(A) \times P(B)$$

Bayes' theorem:

$$P(A \text{ given } B) = P(B \text{ given } A) \times P(A) / P(B)$$

Exam Checklist

Question asks...	Do this
A OR B	Use addition rule; check overlap
A AND B	Use multiplication rule
A given B	Use conditional probability
At least one	Consider complement of none
Without replacement	Update counts after first selection
With replacement	Return item; probabilities reset
Independent	Multiply separate probabilities for AND
Mutually exclusive	Set overlap to zero
Reverse conditional	Consider Bayes' theorem
Not A	Use complement

Final check: do not choose a formula only because you see the word 'OR' or 'AND'. Identify the event structure first, then determine whether there is overlap, dependence, a condition, or a complement.

17. References

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- [OpenStax — The Addition Rule for Probability](#)
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